

SAGAR KUMAR JHA

Software Engineer | Systems & Developer Tools

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[GitHub](#) | [LinkedIn](#) | [LeetCode](#) | [Portfolio](#)

EDUCATION

Uttaranchal University

Master of Computer Application (MCA), Computer Science

India

July 2026 - Present

Indira Gandhi National Open University

Bachelor of Computer Application (BCA), Computer Science

India

July 2019 - July 2024

EXPERIENCE

Software Developer — Neo Capital

Jan 2026

- Built the dashboard UI, owning frontend architecture and implementation using TypeScript, React, and React Router Framework.
- Implemented JWT and session-based authentication with frontend interceptors for authenticated API requests and token/session handling.

Freelance Software Developer — SPG Prints

December 2025 - Jan 2026

- Developed responsive frontend interfaces and interactive animations for the client website using React and TypeScript.
- Collaborated with another developer through Git-based workflows and contributed features through development and deployment.

SKILLS

Programming Languages: C++, TypeScript, JavaScript, Java, SQL

Libraries/Frameworks: React, Electron, Node.js, Vite, Tailwind CSS

Tools / Platforms: Git, GitHub, GitHub Actions, VS Code, npm, CMake, POSIX/Win32

Databases: PostgreSQL, MongoDB

PROJECTS

MiniGit — Git-Compatible Version Control System | [Link](#) C++20, CMake, OpenSSL, zlib, libcurl, POSIX/Win32

- Built Git-compatible storage using SHA-256, zlib compression and binary packfiles, reducing snapshot storage overhead by 60%+.
- Implemented Myers O(ND) diff and LCA-based 3-way merge, reaching 1.29M+ ops/sec in benchmarks.
- Added Git Smart HTTP/HTTPS with pkt-line framing, packfile streaming, and push/pull/fetch support.
- Added rebase, bisect, worktrees and submodules, backed by 18+ test suites and cross-platform CI/CD for Windows, Linux and macOS.

CodeShelf — Cross-Platform Developer Knowledge System | [Link](#) TypeScript, React 19, Electron, Node.js, Vite, Tailwind CSS · [GitHub](#)

- Built an offline-first developer knowledge system across Web, Desktop and VS Code, sharing core logic through a npm monorepo.
- Designed a unified snippet organization system with user-defined folders, technology tags, and extension normalization across file types.
- Added semantic snippet search and technology normalization using Jaccard similarity, with queries completing in ~1.1 ms.
- Built Web Streams compression pipelines with real-time metrics and automated GitHub Actions releases for Windows, macOS and Linux.

ACHIEVEMENTS

- LeetCode Guardian 2382 Contest Rating Top 0.36% globally; solved 715+ problems across algorithms and data structures, including 149 Hard and 361 Medium
- 15+ GitHub Stars across open-source systems projects, including MiniGit and CodeShelf